

Alexandre Autran

Normalien in Machine Learning and Mathematics

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Presentation

I am a machine learning and mathematics student at the Écoles Normales Supérieures (ENS) in Paris-Saclay and Rennes. My research interests are centered on probabilistic and geometric methods for machine learning, high-dimensional statistics, density analysis, optimal transport, graph-based learning, and statistical learning theory.

Education

2024–2027	ENS Diploma, Department of Mathematics Admitted to the ENS through a competitive entrance examination.
2026–2027	École Normale Supérieure Paris-Saclay , Paris, France Research year in machine learning and applied mathematics. Final year of the ENS Diploma (research track at ENS de Rennes).
2024–2027	École Normale Supérieure de Rennes , Rennes, France MSc in Mathematics (year 2). Highest honors (17/20), first semester. MSc in Mathematics (year 1). Highest honors (17.6/20). Rank: 5th/60+.
2024–2025	École Normale Supérieure de Rennes , Rennes, France BSc in Mathematics (final year), Computer Science minor. High honors (15.4/20).
2022–2024	University of Rennes , Rennes, France BSc in Mathematics (years 1 and 2), Computer Science minor. Selective Défi track. Highest honors (18.5/20 and 18/20). Rank: 1st-2nd/100+.

Research Experiences

Aug 2026–	Machine Learning Researcher High-Dimensional Density Analysis Using Local Unimodality. Supervised by Argyris Kalogeratos at Centre Borelli, ENS Paris-Saclay, France.
May–Aug 2026	MSc Research Internship Thermochromic Modeling of the Emissivity of Oxide Nanocomposites. [Report] [Presentation] Supervised by Nguyen Ngoc Duy and Amaury Baret at the University of Liège, Belgium.
Apr–Jun 2026	MSc Research Internship Numerical Simulation and Optimal Control of Spatial Lotka–Volterra Systems. [Report] Supervised by Martin J. Gander and Bastien Chaudet-Dumas at the University of Geneva, Switzerland.
Apr–May 2026	Machine Learning Projects Cardiovascular Disease Prediction Using Statistical Learning and Neural Network Models. [Report] Air Pollution Prediction from Environmental Data. [Report] Studies carried out as part of the MSc course <i>Statistical Learning</i> (prof. Myriam Maumy).
Jan 2026	Statistical Reading Group Supervised Statistical Learning: Prediction by Empirical Risk Minimization. [Presentation] Supervised by Magalie Fromont at ENS de Rennes, France.
Nov–Dec 2025	MSc Research Seminar Point Process and Stein’s Method. [Report] [Presentation] Supervised by Jean-Christophe Breton at the University of Rennes, France.
Oct–Nov 2025	Algebra Reading Group Elimination Theory and Extension. [Presentation] Supervised by Jade Nardi at ENS de Rennes, France.

Jun–Aug 2025	BSc Research Internship Lattice-Based Post-Quantum Cryptography. [Report] [Presentation] Supervised by Pierre-Alain Fouque and Aurore Guillevic at Inria, France.
May–Jun 2025	BSc Research Internship Algebraic Counting of the Real Roots of Polynomials. [Report] [Presentation] Supervised by Marie-Françoise Roy at the University of Rennes, France.
Feb–Mar 2025	Directed Research Reading Geometry of Numbers. [Report] [Presentation] Supervised by Juliette Bavard at the ENS de Rennes, France.
Feb–Mar 2024	BSc Research Internship Topological Study of Matrix Groups. [Report] Supervised by Françoise Dal’Bo and Jérémy Bettinger at the University of Rennes, France.
May–Jun 2023	BSc Research Internship Introduction to Hyperbolic Geometry and Fuchsian Groups. [Report] [Presentation] Supervised by Frank Loray at the University of Rennes, France.

Publications

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- [4] **A. Autran**, "Mean-Field Fluctuations under Emergent α -Stable Common Noise", HAL preprint, 2026. [\[Paper\]](#)
 - [3] **A. Autran**, "Second-Order Asymptotics of Emergent α -Stable Common Noise", HAL preprint, 2026. [\[Paper\]](#)
 - [2] **A. Autran**, "Optimization with an Exact Discrete Adjoint for a Diffusive Lotka–Volterra Harvesting Problem", HAL preprint, 2026. [\[Paper\]](#)
 - [1] **A. Autran**, D. Munteanu, and J.-L. Autran, "Multi-Dimensional Charge Diffusion–Collection Model in Semiconductor Devices Subjected to Single Ionizing Particles", *Applied Sciences*, vol. 16, no. 11, Art. no. 5551, 2026. [\[Paper\]](#)

Other Activities

2024–2025	ENS Oral Examinations Supervision of 2nd year BSc students in preparation for the oral entrance examinations in mathematics of the Écoles Normales Supérieures: Ulm, Paris-Saclay, Rennes and Lyon. [Resources]
2022–2025	Teaching in Mathematics Tutoring for final-year high school students of the Lycée Chateaubriand in Rennes, and for first and second years BSc students of the University of Rennes. [Resources]
2023–2024	BSc Student Representative Representative on the council of the Mathematics Department of the University of Rennes.
Before 2022	High School Competitions Concours Général in Mathematics, Physics, Philosophy, and National High School Mathematics Olympiad at Lycée Thiers in Marseille.

Computer Skills

Programming: Python, C/C++, Java, MATLAB, R. **Python / ML:** NumPy, pandas, scikit-learn, PyTorch.

Languages

French: native language. **English:** professional working proficiency.

Sports and Interests

Sports: running, cycling, strength training. **Interests:** reading and writing, in fantasy and science fiction literature; chess.

Academic Reference

François Bolley [\[Web page\]](#)

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